EEL 5764: Computer Architecture

Lecture 3: Market Segments, Technology Trends, and the Scaling Crisis

Sandip Ray

Warren B. Nelms Endowed Professor

Department of Electrical & Computer Engineering

University of Florida

Required Reading: H&P Sections 1.3 & 1.4

Physics vs. Architecture: Why Technology Drives Innovation

THE DEVICE PHYSICS BOUNDARY

- Transistors as Building Blocks: Physical gates, leakage currents, and wiring delay dictate microarchitectural feasibility.
- The Scaling Imperative: For decades, smaller transistors automatically meant faster circuits and less power per function.
- The Abstraction Gap: Hardware architects translate physical gate characteristics into programmable execution primitives.
- The Hard Limits: When device-level physics hits a wall, the architecture must change to maintain system-level scaling.

THE ARCHITECTURAL RESPONSIBILITY

- Bridging Software & Hardware: Creating abstractions (like ISAs) that mask complexity while enabling optimization.
- Designing under Physical Constraints: Managing area budgets, cooling limits, and timing closures on the physical silicon.
- Measuring Quantitative Tradeoffs: Proving that allocating more transistors to a mechanism (e.g., caches) pays off.
- Historical Perspective: Traditional single-core performance scaled automatically with transistor shrink—until the walls hit.

Market Segment 1: Ultra-Low Power & IoT

Critical Constraints & Envelopes

- Microwatt Envelopes: Often operates under strict power budgets, sometimes utilizing ambient energy-harvesting technologies.
- Extreme Cost Sensitivity: Target prices can be less than 50 cents per microchip.
- Sensing Interface integration: Analog-to-digital converters and wireless links embedded on a single silicon die.

Example Domains

- Medical Implants: Cardiac pacemakers, neural interfaces, and glucose sensors requiring absolute reliability and multi-decade battery life.
- Smart Dust & RFID: Tiny ambient sensors deployed in structural materials to monitor temperature and physical stress.
- Industrial IoT Nodes: Remote environmental trackers operating on small coin-cell batteries.

Market Segment 2: Mobile & Client Devices

Design Drivers & Limits

- Passive Thermal Dissipation: No physical fans. Power is capped to prevent the chassis temperature from exceeding comfortable human limits (~45°C).
- Battery Conservation: Idle leakage must be practically zero to ensure long battery standby times.
- High Burst Performance: Demands highly responsive, multi-core performance for short task durations.

Primary Platforms

- Modern Smartphones: Heterogeneous multi-core SoCs running high-end graphics and real-time networking workloads concurrently.
- Smartwatches & Wearables: Compact packaging with strict area footprints and thermal limits.
- AR / VR Headsets: Real-time, ultra-low latency sensory data processing demanding highly specialized microarchitectures.

Market Segment 3: Warehouse-Scale & AI Clusters

The Industrial Optimization space

- Megawatt Power budgets: Operating costs are heavily dictated by energy bills, driving maximum optimization of performance-per-watt metrics.
- Active cooling infrastructure: Liquid loops, massive heat exchangers, and fan racks are deployed to handle immense thermal outputs.
- Extreme Memory Bandwidth: Deploys stacked High-Bandwidth Memory (HBM) to avoid the physical limits of standard memory links.

Computing Infrastructure

- Warehouse-Scale Centers: Tens of thousands of linked nodes running cloud services, databases, and enterprise virtual machines.
- AI Cluster accelerators: Specialized tensor calculation engines (e.g., Google TPU, NVIDIA H100) running massive AI model training workloads.
- High-Performance Supercomputing (HPC): Dense clusters running complex physical and weather simulations.

Analyzing Power: The Dynamic Power Equation

DYNAMIC TRANSISTOR SWITCHING POWER

- The Dynamic Power Formula (Charging and discharging internal load capacitances):

$$P_{dynamic} = C * V^2 * f * alpha$$

C: Capacitive load of the circuit logic.

V: Supply voltage (dominant squared factor!).

f: Operating frequency (clock speed).

alpha: Activity factor (average frequency of logic state transitions).

- Capping Clock Speeds: Once voltage scaling slowed substantially, continued frequency increases became increasingly expensive in power, motivating limits on frequency growth and a broader range of architectural power-management techniques.

ENERGY VS. POWER TRADEOFF

- Power represents rate of work (critical for thermal design, fan constraints, battery size). Unit: Watts.
- Energy represents total work done over time (critical for battery lifetime). Unit: Joules.

Energy = Power * Execution Time

- Optimizing for Energy: Sometimes, running a processor at a higher voltage/frequency for a shorter period ("Race to Sleep") consumes less energy than running at low speed for a long time.

Moore's Law: The Economic Engine of Silicon Scaling

WHAT GORDON MOORE ACTUALLY SAID

- The Observation (1965): Transistor density on an integrated circuit doubles roughly every 1.5 to 2 years.
- An Economic Target, Not Physics: Moore's Law is a business pace that drives lithography tool research and investment.
- Staggering Historical Scale: Progressed from thousands of transistors in early microprocessors to over 100 billion transistors on a single monolithic die today (and even more using advanced packaging).
- Modern Challenges: Extreme ultraviolet lithography (EUV), multi-patterning, and escalating non-recurring engineering (NRE) costs make physical transistor scaling increasingly difficult and expensive.

ARCHITECTURAL IMPLICATION

If the density doubles every two years, an architecture that was impossible five years ago becomes standard today, and an obsolete design gets crushed under exponential physics.

- 1971 (Intel 4004): 2,300 transistors
- 2026 (Modern SoCs): ~100+ billion transistors
- Growth Factor: >43,000,000x scaling over 55 years!

Dennard Scaling: The Golden Age of Power-Free Performance

THE SCALING FORMULATION

- Robert Dennard's Principle (1974): As transistors shrink in physical dimensions, their electrical power density remains completely flat.
- The Mechanics: If we scale physical dimensions (width, height, thickness) down by a factor of 1/K, we also scale the voltage (V) down by 1/K, and the capacitance (C) down by 1/K.
- The Consequence: The switching delay scales down by 1/K, allowing us to safely scale the clock frequency (f) up by K.
- The Result: Overall performance increased by ~K, while total chip power consumption stayed constant! We got faster chips with zero thermal penalty.

Parameter	Dimensions	Scaling Factor
Dimension (W, L, t_ox)	Width, Length, Oxide Thk	1 / K
Voltage (V)	Supply Voltage	1 / K
Current (I)	Device Current	1 / K
Capacitance (C)	Gate Capacitance	1 / K
Delay (t)	Switching Delay	1 / K
Frequency (f)	Clock Frequency	K

The Voltage and Leakage Limits: Why Dennard Scaling Collapsed

BEFORE 2005: THE GOLDEN ERA

- Voltage Scaled Down: We safely reduced supply voltages from 5V to 3.3V, 2.5V, 1.8V, and down to 1.2V.
- Static Power Negligible: Current only flowed when transistors switched states; standby leakage was near zero.
- Performance Chased Clock Frequency: Microprocessors increased frequencies from 10 MHz to 3 GHz with flat power density.

AFTER 2005: THE COLLAPSE

Physical Scaling Limits: As transistor dimensions shrank, supply and threshold voltages could no longer scale proportionally with dimensions.

- The Voltage Floor: Supply voltage could no longer scale proportionally with transistor dimensions without unacceptable leakage, variability, and reliability penalties.
- Multiple Physical Barriers: Gate-oxide tunneling leakage, short-channel effects, process variability, and reliability limits combined to dominate power budgets (Pstatic = Ileakage * Vsupply).

Dennard Scaling Ended: With voltage capped, scaling physical size down while keeping frequency high caused dynamic and static power density to skyrocket.

The Power Wall & The Rise of 'Dark Silicon'

THERMAL DISCIPLINE & THE COLD WALL

- The Power Wall: Power and thermal constraints made continued voltage/frequency scaling increasingly impractical.
- Thermal Density Limits: Shrinking transistor size without scaling voltage increases heat density, making advanced active cooling expensive or impractical.
- The Dark Silicon Phenomenon: As transistor density continues to increase faster than the power budget, not all available circuitry can operate simultaneously at maximum activity and frequency.
- Architectural Response: Rather than making general-purpose cores larger, dark silicon drives us to build highly specialized, heterogeneous accelerators.



The Instruction-Level Parallelism (ILP) Wall

THE ILP WALL: DIMINISHING RETURNS

- Pipelining and Superscalar Limits: Dynamic pipelines try to automatically find independent instructions within a single thread of execution.
- The Complexity Explosion: To extract minor speedups, architects built massive branch prediction tables, deeper pipelines, and huge out-of-order instruction scheduling windows.
- The Diminishing Returns: Many structures needed for wide out-of-order execution scale superlinearly in complexity, area, and power, while available ILP exhibits diminishing returns.
- The Collapse: Single-threaded code reaches a natural execution dependency limit (data hazards, pointer aliasing, control hazards), slowing the historical single-thread performance trend.

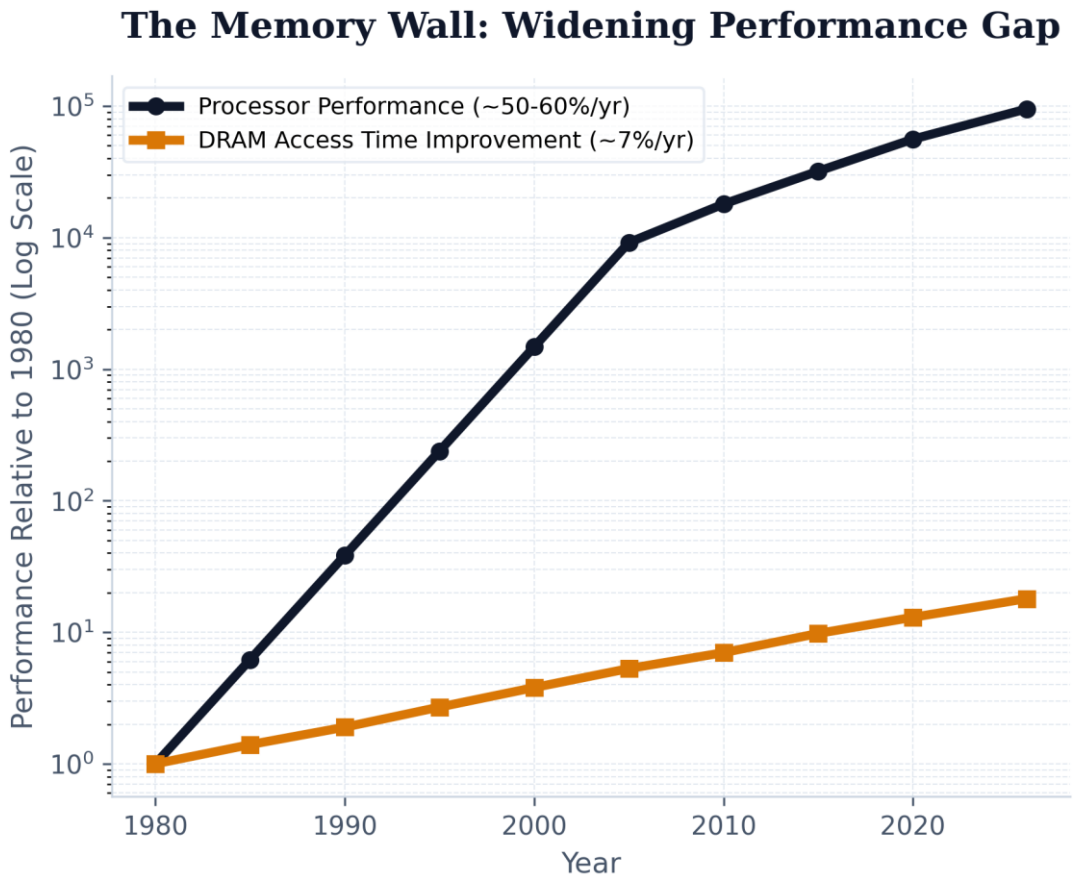
THE HARDWARE COST OF ILP

- Branch Predictors: Complex algorithms (gshare, TAGE) consuming thousands of SRAM bits to predict branch targets.
- Reorder Buffers (ROB): Schedulers must track 128+ active, in-flight instructions to resolve dependencies in flight.
- Register Renaming Tables: Multi-port SRAM arrays to decouple temporary registers (WAR/WAW hazards).
- The Result: Extremely high static and dynamic power overhead for very little gain on standard general-purpose codes.

The Memory Wall: CPU Speed vs. DRAM Latency

THE SPREADING LATENCY GAP

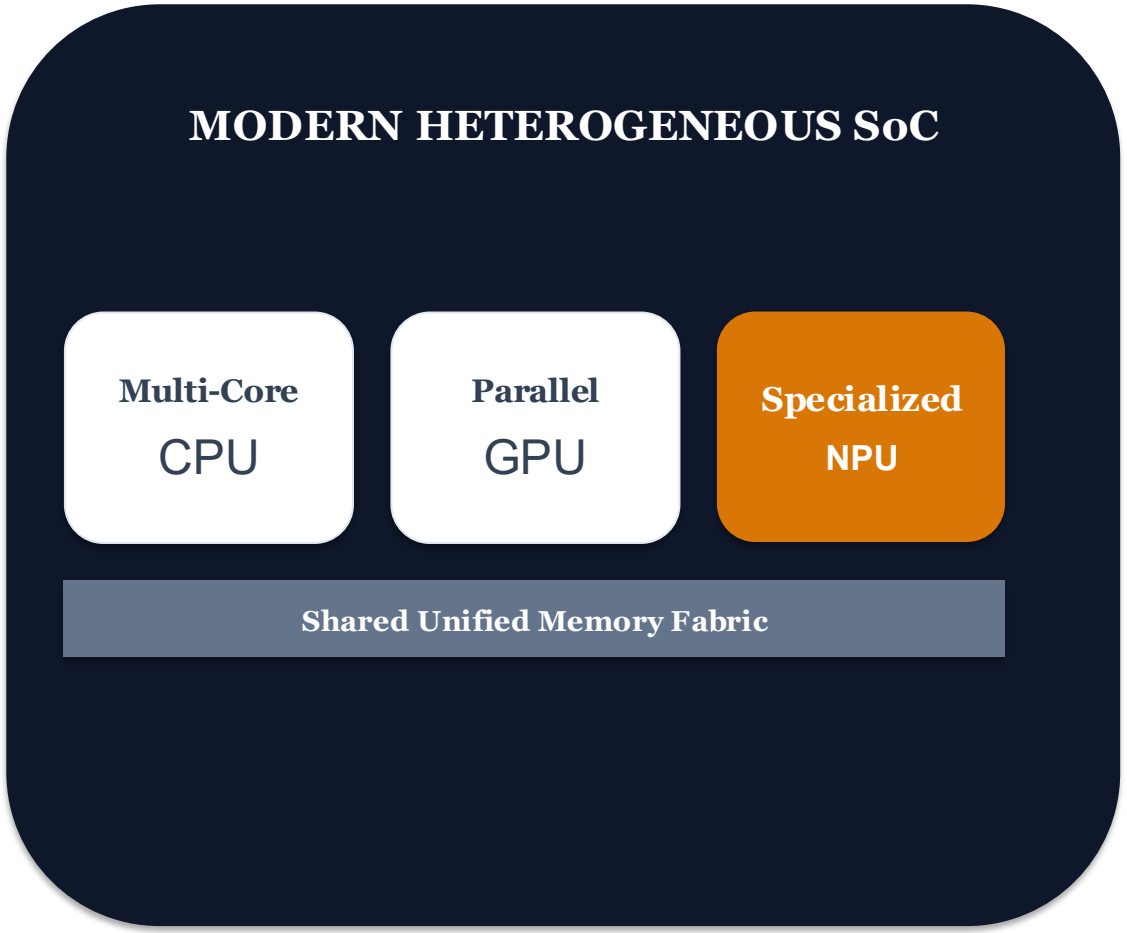
- The Memory Wall: The exponentially growing performance and latency gap between processor execution and main memory access.
- Historical Trends: Historically, single-core execution performance grew at ~50–60% per year, while DRAM memory latency only improved at ~7% per year, creating an exponentially widening gap.
- The Starvation: A CPU core running at 3.5 GHz can execute multiple instructions per nanosecond, but a main memory access (DDR DRAM) takes ~50–100 nanoseconds, stalling execution for hundreds of cycles.
- Caches as Lifeboats: We manage the Memory Wall by integrating multi-level, massive cache hierarchies (L1, L2, L3) directly on the CPU die.



The Structural Pivot: Multicore and Specialization

THE NEW PARADIGM

- Capping Single-Thread Speeds: Since we can no longer scale single-core frequency or ILP, the industry was forced to pivot.
- The Multicore Strategy (Spatial Scaling): Rather than building one massive, power-hungry core, we share the power budget across multiple simpler, parallel cores on a single die.
- The Specialization Strategy (Domain-Specific): Building custom hardware units (accelerators) designed to do one class of math extremely efficiently.
- The Efficiency: Accelerators (like GPUs, NPUs, video decoders) eliminate general-purpose instruction fetching and scheduling overhead, potentially yielding order-of-magnitude improvements for suitable workloads, trading generality for efficiency.



Discussion: Allocating Dark Silicon

If you are a lead System-on-Chip (SoC) architect designing a modern processor:

How do you leverage billions of 'dark' (or underutilized) transistors to add value to the user, without violating your fixed thermal/power envelope?

1. Dedicated Accelerator Blocks: Can they be completely power-gated when idle?
2. Cache Size Tradeoffs: Can additional cache capacity reduce expensive memory traffic enough to justify its own area and energy cost?
3. Spatial vs. Temporal Multi-Core Scheduling: Can we alternate hot active cores?